LDR BASED PEOPLE COUNTER

1. Introduction

This project aims to design a simple **automatic people counter** system that can count the number of people entering or leaving a room.
It uses a **Light Dependent Resistor (LDR)** as a sensing element, **LM358 Op-Amp** as a comparator, and a **CD4026 counter IC** to display the count on a **7-segment common cathode display**.

Whenever a person passes through the entrance, the LDR detects a change in light intensity because person cuts the light source falling on the LDR which is **LASER LIGHT**, triggering a count increase or decrease. The system is useful in places like classrooms, offices, libraries, or auditoriums where knowing the number of people helps manage space and energy efficiently.It also comprise with a 5V buzzer which beeps every time number changes on 7 segment display.

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2. Objective

To design and implement an **automatic digital people counter** that counts individuals entering or exiting a room using light interruption detection and displays the count digitally on a 7-segment common cathode display.

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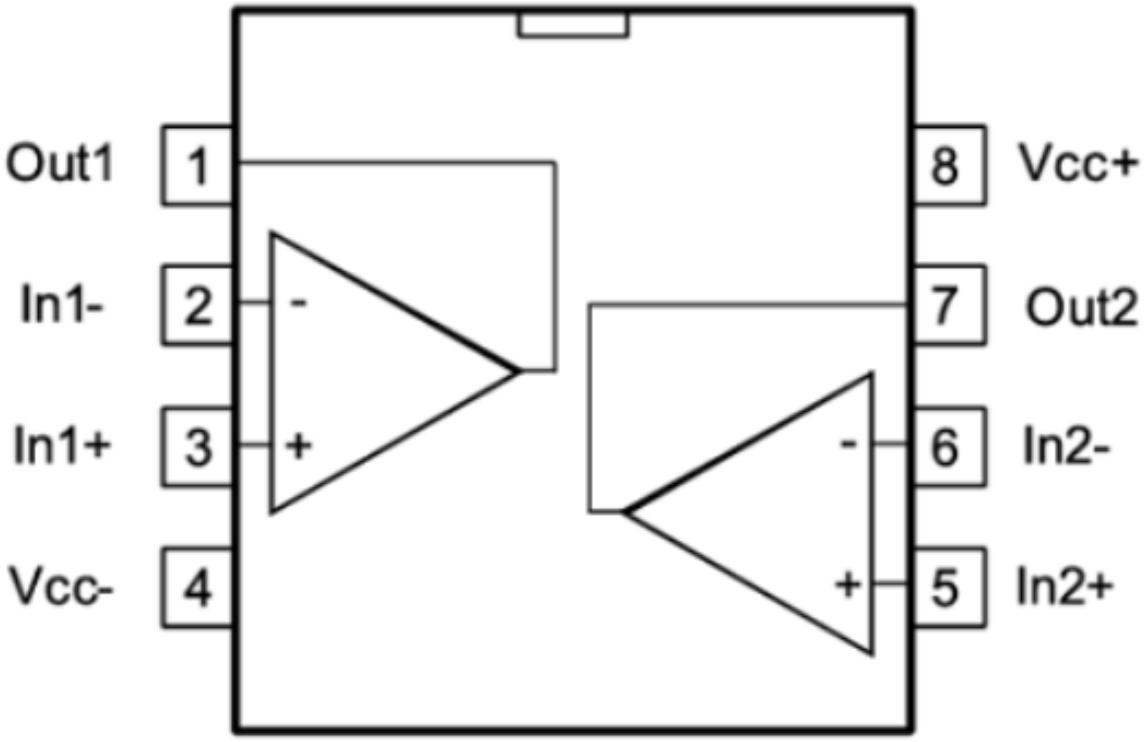
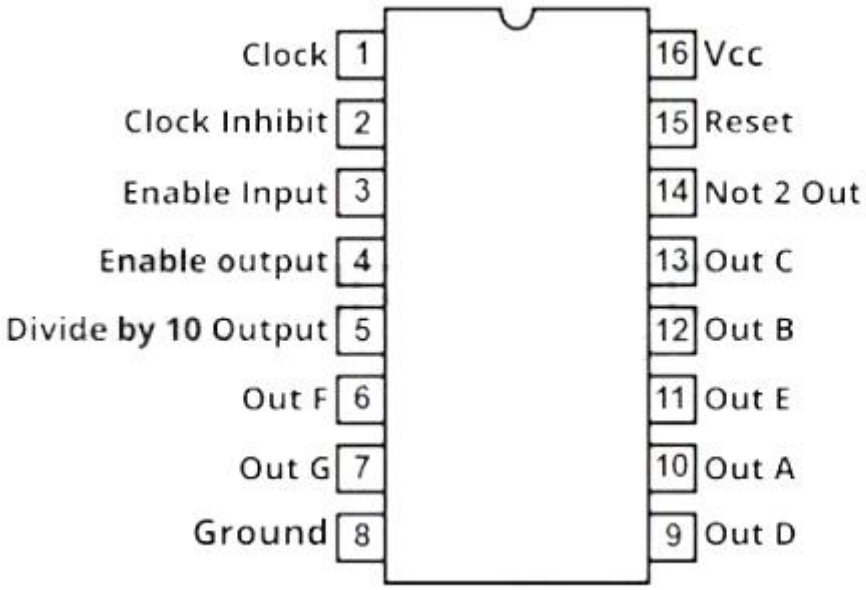
3. Components Required

S.No	Component	Specification	Quantity
1	IC LM358	Dual Operational Amplifier	1
2	IC CD4026	Decade Counter with 7-Segment Driver	1
3	LDR	Light Dependent Resistor	1
4	7-Segment Display	Common Cathode	1
5	Resistors	1kΩ	3
6	Potentiometer	10kΩ	1
7	Capacitor	100nF, 10μF	1
8	Transistor	BC547	1
9	Push Button	Optional (for reset or triggering)	1
10	Power Supply	5V–9V DC	1
11	Laser Light Source	—	1

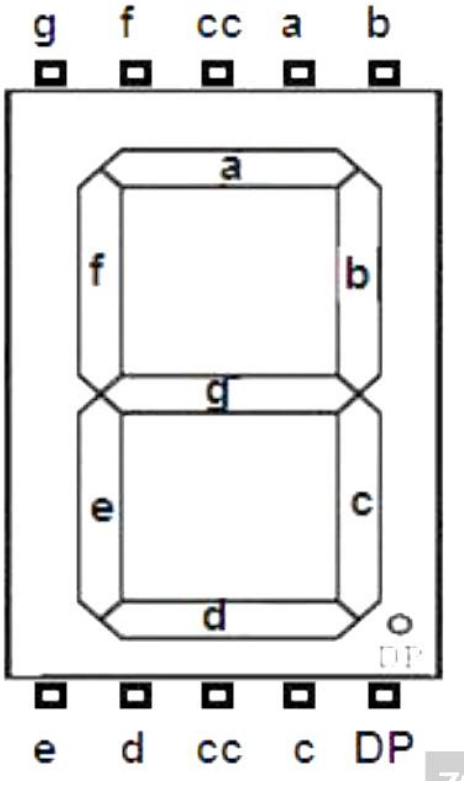
4(a). LM358 IC Pin Diagram



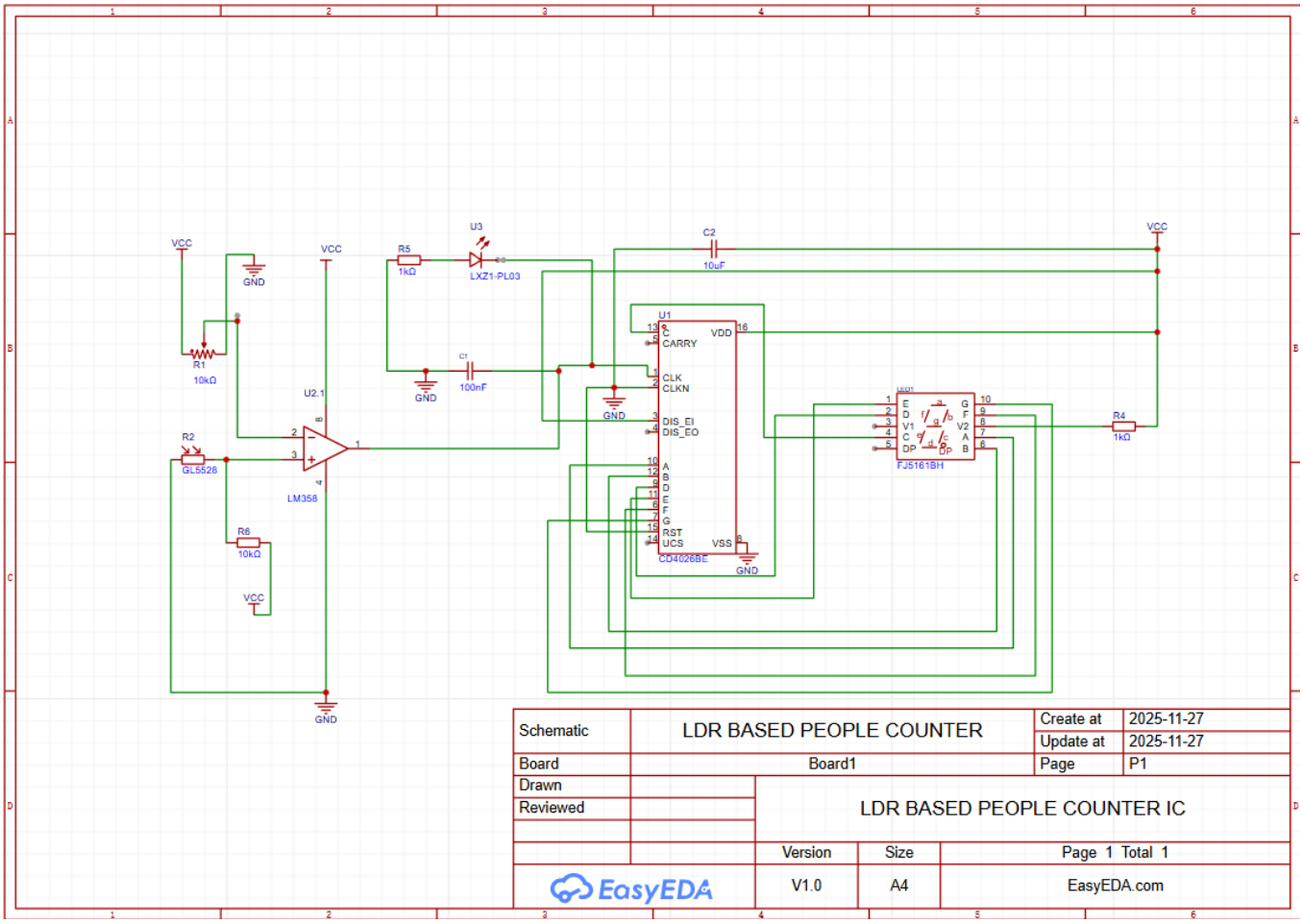
4(b). CD4026 IC Pin Diagram



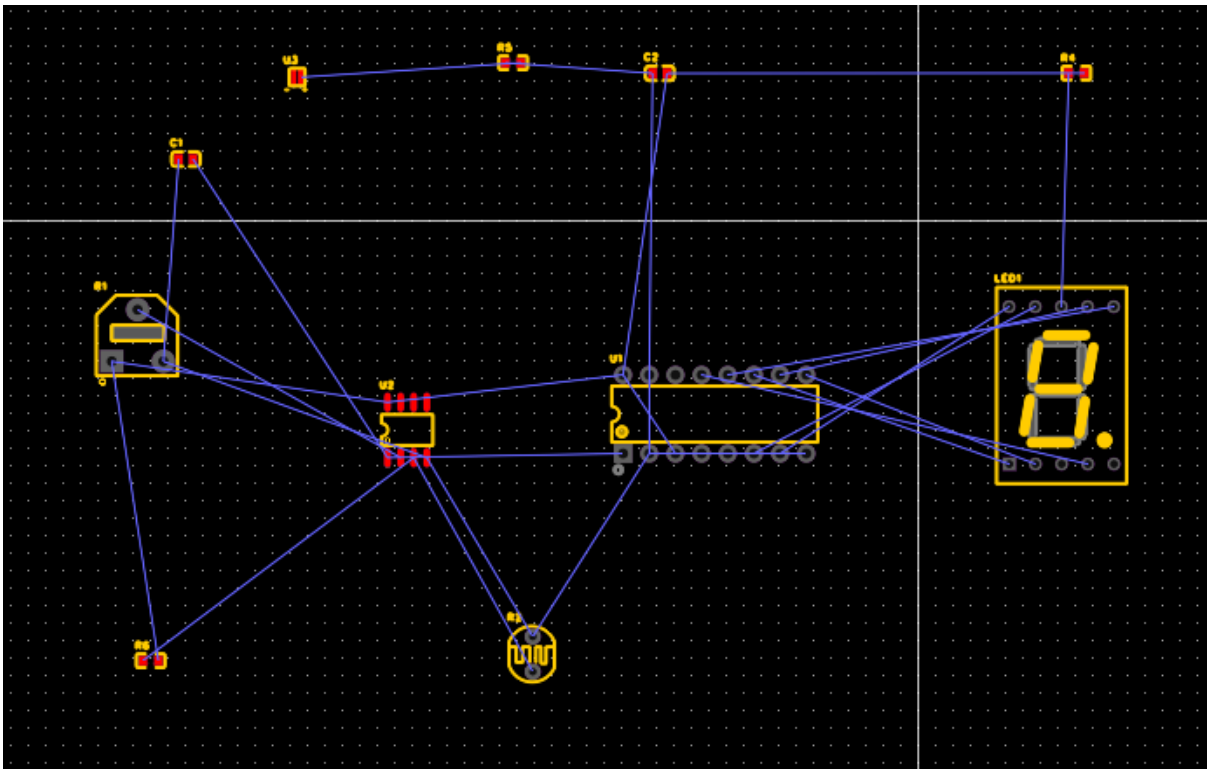
4(c). 7 Segment Display Common Cathode Pin Diagram



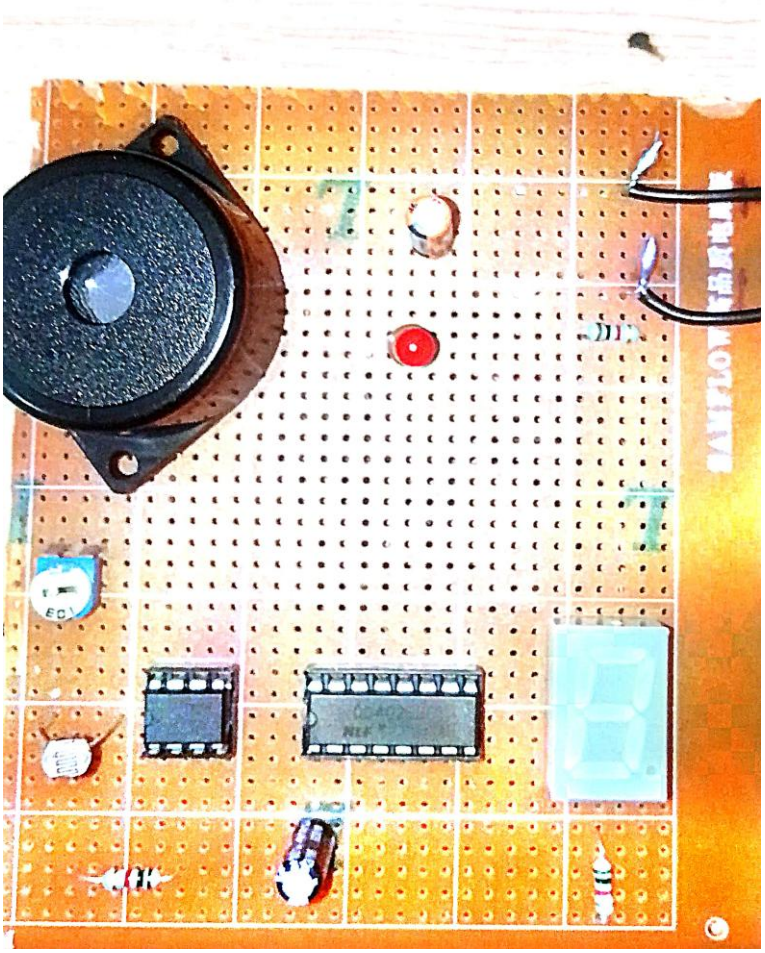
4(d). Schematic Diagram



4(d). PCB Implementation



4(e). Zero PCB Implementation



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5. Circuit Description

(a) Sensor Section

- The **LDR** and a **fixed resistor** form a **voltage divider** circuit.
- When light falls on the LDR, its resistance decreases, causing the voltage at the junction to change.
- When a person passes in front of the sensor and blocks the light, the resistance of the LDR increases sharply, changing the voltage level.

(b) Comparator Section (LM358)

- The LM358 compares the voltage from the LDR divider at the **non-inverting input** with a **reference voltage** (set by a potentiometer at the **inverting input**).
- When the light beam is interrupted, the output of the comparator switches state (from LOW to HIGH or vice versa).
- This sharp transition acts as a **trigger pulse** for the CD4026 counter.

(c) Counter and Display Section

- The output of the LM358 is connected to the **clock input (pin 1)** of the **CD4026 IC**.
- The CD4026 counts each clock pulse and automatically drives the **7-segment display** to show the number.
- Each pulse corresponds to one person passing through the entrance.
- A **reset button** can be used to reset the count to zero by connecting it to the **reset pin (pin 15)** of CD4026.

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6. Working Principle

1. The LDR continuously receives light from a fixed light source (like an LED or laser).
2. When a person passes between the source and LDR, the light is momentarily blocked.
3. This interruption changes the voltage across the LDR, which is detected by the LM358 comparator.
4. The comparator output goes HIGH momentarily, producing a pulse.
5. This pulse is fed into the clock input of the CD4026, which increases the count by one and updates the 7-segment display.
6. The circuit resets and waits for the next interruption.

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7. Applications

- Automatic **entry/exit counters** in rooms, halls, and buses.
- **Crowd management** and **energy-saving** systems (e.g., turning off lights when the room is empty).
- Useful for **library seat counters** or **office occupancy monitoring**.
- Can be integrated with microcontrollers for smart automation.

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8. Advantages

- Fully **automatic and contactless** counting.
- Uses **low-cost, easily available** components.
- Can run on a small battery.
- Simple design, easy to expand for dual-direction counting.

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9. Limitations

- Sensitive to **ambient light variations** — requires a stable light source.
- Works reliably only when **one person passes at a time**.
- Doesn't differentiate between **entry and exit** unless dual sensors are used.
- No memory — loses count when power is turned off (unless memory IC or MCU is added).

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10. Conclusion

This project successfully demonstrates a simple and efficient **people counting system** using an **LDR sensor**, **LM358 comparator**, and **CD4026 counter driver**.
The circuit works on the basic principle of **light interruption detection** and provides a digital count output. It serves as a foundational project to understand **sensors, comparators, digital counters, and display driving** — all key concepts in electronics and automation.